



Open Cosmos Hyper-500 L1C: Quality Processes Review

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1 Introduction

1.1 Scope

This document summarises a Data Product Quality Processes Review for OpenCosmos Hyper-500 L1C as part of the EODH Data Product Quality Process [AD-1]. The purpose is to demonstrate that the product has been developed and generated following processes that adhere to best practices.

To achieve this, the EODH Data Product QA Process adheres to the ESA/NASA EO Product QA Framework [RD-1]. This framework defines requirements for a variety of aspects of Data Product quality – awarding grades of *Basic*, *Good*, *Excellent*, or *Ideal*. Such assessments are reported with a maturity matrix – this is presented in Section 2. The rest of the document provides the underpinning evidence justifying the awarded grades. This assessment has been completed by EODH QA Service on behalf of the Data Product provider.

The result of this evaluation, referred to as a *QA Check Result*, will be added as an annotation to the EODH Catalogue for user interaction.

1.2 Changes Log

Issue	Date	Description
1.0	03/03/2026	Draft for pre-publication review.

1.3 Applicable Documents

[AD-1] EODH QA Service, ‘Data Product Quality Assurance Process’, 2025.

1.4 Reference Documents

[RD-1]	Hunt et al., ‘Earth Observation Mission Quality Assessment Framework’ v2.2, EDAP Project Document, 2021. Available Online .
[RD-2]	OpenApp Hyper-500 Data Specification. See: https://app.open-cosmos.com/help/data-catalog/products/hyper-500
[RD-3]	Wilkinson, M., Dumontier, M., Aalbersberg, I. et al. The FAIR Guiding Principles for scientific data management and stewardship. Sci Data 3, 160018 (2016); https://doi.org/10.1038/sdata.2016.18
[RD-4]	https://cogeo.org/

1.5 Acronyms & Abbreviations

EDAP	Earthnet Data Assessment Project
EO	Earth Observation
EODH	Earth Observation Data Hub
ESA	European Space Agency
NASA	National Aeronautics and Space Administration
NCEO	National Centre for Earth Observation
NPL	National Physical Laboratory

QA

Quality Assurance

2 Review Summary – Product Maturity Matrix

The *Data Product Quality Processes Review* is aimed at demonstrating that a given Data Product is developed and generated following processes that adhere to best practices.

To achieve this, the EODH *Data Product QA Process* adheres to the ESA/NASA EO Product QA Framework [RD-1]. Such assessments are reported in a maturity matrix style, where each box represents an area of assessment for which underpinning requirements are defined – the box colour represents the achieved grade, *Basic*, *Good*, *Excellent*, or *Ideal*. Additionally:

- *Not Assessed* means this aspect of data quality was not assessed at this time.
- *Not Assessable* means the assessment of this aspect of data quality was not possible due to a lack of available evidence.
- The padlock symbol means that the supporting evidence reviewed is not publicly available.

The *Data Product Quality Processes Review* for OpenCosmos Hyper-500 L1C is presented in Figure 1. The rest of the document provides the underpinning evidence justifying the awarded grades, as follows:

- Section 3 provides the evidence for the Product Information column.
- Section 4 provides the evidence for the Metrology column.
- Section 5 provides the evidence for the Product Generation column.

Data Product Quality Processes Review			Key
Product Information	Metrology	Product Generation	
Product Details	Sensor Calibration & Characterisation	Calibration Algorithm	Not Assessed
Availability & Accessibility	Geometric Calibration & Characterisation	Geometric Processing	Not Assessable
Product Format, Flags & Metadata	Metrological Traceability Documentation		Basic
User Documentation	Uncertainty Characterisation		Good
	Ancillary Data		Excellent
			Ideal
			Not Public

Figure 1 - Data Product Quality Processes Review Maturity Matrix for OpenCosmos Hyper-500 L1C

3 Product Information

Product Details																							
Grade: Good																							
Justification	Some information supplied in the OpenApp Hyper-500 Data Specification [RD-2], however some important pieces of information missing, including stated measurement quality, mission lifetime, temporal resolution and orbital information for the Hammer and Acc1 spacecraft.																						
Product Name	OpenCosmos Hyper-500 L1C																						
Sensor Name	Hyper-500																						
Sensor Type	Optical – Hyperspectral (32 channels) in the visible and NIR range.																						
Mission Type	Constellation – 2 satellites (Hammer & Acc1)																						
Mission Orbit	Sun-Synchronous Orbit (SSO) at 14:00 hours Local Time Descending Node (LTDN).																						
Product Version Number	Not stated.																						
Product ID	The file names for Hyper-500 data products follow a standard convention as defined on the OpenApp Data Specification: <table border="1"> <thead> <tr> <th>Field</th><th>Description</th><th>Example</th></tr> </thead> <tbody> <tr> <td>Mission</td><td>Satellite/mission name</td><td>HAMMER</td></tr> <tr> <td>Level</td><td>Processing level</td><td>L1A (also L0, L1B, L1C)</td></tr> <tr> <td>Session</td><td>9-digit internal session ID</td><td>000000000</td></tr> <tr> <td>Start</td><td>Acquisition start (UTC, YYYYMMDDhhmmss)</td><td>20260722221802</td></tr> <tr> <td>End</td><td>Acquisition end (UTC, YYYYMMDDhhmmss)</td><td>20260722221812</td></tr> <tr> <td>CRC</td><td>Checksum (CRC32-C), unique identifier</td><td>1FA726CB</td></tr> </tbody> </table> e.g. HAMMER_L1A_00000000X_20260722221802_20260722221812_1FA726CB		Field	Description	Example	Mission	Satellite/mission name	HAMMER	Level	Processing level	L1A (also L0, L1B, L1C)	Session	9-digit internal session ID	000000000	Start	Acquisition start (UTC, YYYYMMDDhhmmss)	20260722221802	End	Acquisition end (UTC, YYYYMMDDhhmmss)	20260722221812	CRC	Checksum (CRC32-C), unique identifier	1FA726CB
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CRC	Checksum (CRC32-C), unique identifier	1FA726CB																					
Processing level of product	Level 1C data products have TOA reflectance units and have been orthorectified with a moderate resolution DEM, resulting in a cartographic product with accurate geolocation.																						
Measured Quantity Name	TOA Reflectance																						
Measured Quantity Units	N/A																						
Stated Measurement Quality	Radiometric Quality: TBD at time of assessment. Geometric Quality: TBD at time of assessment.																						
Spectral Resolution	32 channels in the visible and near infrared spectral range. Band central wavelengths and FWHM stated within RD-2																						
Spatial Resolution	High-resolution Ground Sampling Distance (GSD): 4.75m @ Nadir																						

Spatial Coverage	<i>Hammer:</i> <i>Orbital period: Not stated.</i> <i>Orbit inclination: Not stated.</i> <i>Acc1:</i> <i>Orbital period: Not stated.</i> <i>Orbit inclination: Not stated.</i>
Temporal Resolution	<i>Revisit capacity: Not stated.</i>
Temporal Coverage	<i>Mission lifetime: Not stated.</i> <i>Launches:</i> <i>Hammer: Not stated.</i> <i>Acc1: Not stated.</i>
Point of Contact	<i>sales@open-cosmos.com</i>
Product locator (DOI/URL)	<i>Products available on request via https://www.open-cosmos.com/data-and-services#Contact-Form</i>
Conditions for access and use	<i>Not stated.</i>
Limitations on public access	<i>Not publicly available.</i>
Product Abstract	<i>From [RD-2]:</i> <i>The Hyper-500 product offers high-resolution hyperspectral imagery, capturing 32 bands at under 5m resolution.</i> <i>The hyperspectral payload captures 32 channels in the visible and near infrared spectral range. The imager captures in linescan mode using a wedge filter. The 32 spectral bands central wavelengths can be chosen by the band start row, with the filter width being influenced by the number of dTDI stages in use.</i>

Availability & Accessibility	
Grade: Basic	
Justification	<i>The product information provided, such as Product Locator and Conditions for access and use, meets only some of the FAIR principles.</i> <i>Findable, Accessible, Interoperable and Reusable (FAIR) Principles – see [RD-3].</i>
Compliant with FAIR principles	No
Data Management Plan	Description of data management plan is not available.
Availability Status	Products available on request via https://www.open-cosmos.com/data-and-services#Contact-Form

Product Format, Flags & Metadata				
Grade: Good				
Justification	<i>COG (Cloud Optimized Geotiff) [RD-4], a documented standard file format, is used. The products also include a good set of documented metadata with a self-defined standard naming convention, but data flag information is not made available. Product is not CEOS ARD compliant.</i>			
Product File Format	Products are available in a well-defined file format structure defined at [RD-2]:			
	Title	Key	File Format	Description
	Band data	PAN, HS1, HS2, HS3, etc	COG	Cloud optimised raster data containing band spectral data
	Thumbnail (TCI)	thumbnail	webp	A low resolution true colour composite image for quick visualisation of imagery data, typically used as a preview in the browser
	Metadata and ancillary	metadata	xml and json	to be defined in detail
	Photogrammetric metadata	ancillary	json	Contains both intrinsic (sensor size, pixel size, focal length, principal point, aperture length, reference frame) and extrinsics (position, velocity, angular rates, and attitude) parameters

	<i>Platform metadata</i>	<i>metadata</i>	<i>json</i>	<i>Platform and payload information (firmware & software versions, imager configuration, sensor temperature, among others)</i>
Metadata Conventions	<i>STAC</i>			
Analysis Ready Data?	<i>No</i>			

User Documentation		
Grade: Basic		
Justification	There is no specific user guide, but a Data Specification page is available with high-level information (see [RD-2]). There is no ATBD made publicly available.	
<i>Document</i>	<i>Reference</i>	<i>QA4ECV Compliant</i>
Product User Guide	<i>N/A</i>	<i>No</i>
ATBD	<i>Documentation not made available to users</i>	<i>N/A</i>

4 Metrology

Sensor Calibration & Characterisation	
Grade: Not Assessable	
Justification	<i>Any formal sensor calibration and characterisation documentation (for either pre-flight or post-launch) is not available publicly and therefore it is not possible to deem “fit for purpose” in terms of the mission’s stated performance.</i>
References	

Geometric Calibration & Characterisation	
Grade: Not Assessable	
Justification	<i>Any formal sensor calibration and characterisation documentation (for either pre-flight or post-launch) is not available publicly and therefore it is not possible to deem “fit for purpose” in terms of the mission’s stated performance.</i>
References	

Metrological Traceability Documentation	
Grade: Not Assessable	
Justification	<i>There are no metrology documentation or traceability chain / uncertainty tree diagrams publicly available.</i>
References	

Uncertainty Characterisation	
Grade: Not Assessable	
Justification	<i>Uncertainty information is not publicly available nor is any detailed information regarding the methods used to characterise the uncertainties publicly available.</i>
References	

Ancillary Data	
Grade: Not Assessable	

Justification	<i>There is information on the format of the ancillary data but no description of what specific ancillary data is included with the product and therefore there is neither any detailed information regarding uncertainty characterisation for any of the ancillary data.</i>
References	

5 Product Generation

Calibration Algorithm	
Grade: Not Assessable	
Justification	<i>There is some very high-level information at [RD-2] which outlines the main radiometric corrections that have been carried out from L1A to L1C:</i> <i>Radiometric correction - apply the radiometric coefficients, part of the L1A product assets, to convert L1A DN's into radiance, accounting for dark and flat fields</i> <i>Conversion from TOA radiance to TOA reflectance</i> <i>However, any details on the methods used to evaluate these corrections is not available publicly and therefore it is not possible to deem "fit for purpose" in terms of the mission's stated performance.</i>
References	<i>RD-2</i>

Geometric Processing	
Grade: Not Assessable	
Justification	<i>There is some very high-level information at [RD-2] which outlines the main geometric processes that have been carried out from L1A to L1C:</i> <i>Perform the band alignment of all bands with respect to the reference one (red band)</i> <i>Apply the RPCs, part of the L1A product assets, to georeference the rasters</i> <i>RPC refinement using a global reference image (GRI) and a digital elevation model (DEM)</i> <i>Orthorectification</i> <i>However, any details on the methods used to evaluate these corrections is not available publicly and therefore it is not possible to deem "fit for purpose" in terms of the mission's stated performance.</i>
References	